

Arabic Robot Research — Master Summary

Prepared: July 2026
Hardware baseline: NVIDIA Tesla T4 (~14.9 GB VRAM)
Scope: Combined summary of all deliverables in this folder.

Source report	Topic
ASR_Model_Selection_Report.pdf	Egyptian Arabic speech recognition bake-off
LLM_Model_Selection_Report.pdf	Conversational LLM bake-off
TTS_Model_Selection_Report.pdf	Arabic TTS bake-off
Combined_Families_Summary.pdf	Cross-family stack + VRAM budget
VPS_Provider_Research_Report.pdf	GPU cloud / VPS selection + capacity

1. Bottom line

Production default stack

Family	Winner	Why
ASR	Whisper-Large-v3-Turbo-CT2	Best accuracy/speed/VRAM blend — robot score 89.62 , Acc 68.4% , RTF 0.046 (~22×), VRAM ~2.5 GB
LLM	Nile-Chat-4B	Best turn-taking feel — robot score 95.41 , TTFT 0.96 s , 11.56 tok/s , VRAM ~8.4 GB
TTS	VoiceTut-TTS	Best speed/resources — robot score 90.02 , RTF 0.157 (~6.4×), 82.7 chars/s , VRAM ~3.15 GB

Combined naive peak VRAM ≈ 14.0 GB — fits a T4 tightly; deploy on **≥24 GB** (RTX 4090 class) for production headroom.

Hosting default

RunPod Secure Cloud — RTX 4090 24 GB (~\$0.69/hr). Use Community / Vast.ai for cheap R&D; move live traffic to Secure before client demos.

2. ASR summary

12 models evaluated on a shared Egyptian Arabic clip (100% success).
Robot score weights: **45% accuracy + 30% speed/RTF + 15% VRAM + 10% load**.

Recommended picks

Role	Model	Key metrics
Production default	Whisper-Large-v3-Turbo-CT2	Robot 89.62 , Acc 68.4% , RTF 0.046 , VRAM ~2.5 GB
Best accuracy	Whisper-Large-v3-CT2	Acc 72.92% , WER 0.271 , RTF 0.133 , VRAM ~4.3 GB
Lowest VRAM	Whisper-Small-CT2	VRAM ~1.3 GB , Acc 63.2%
Fastest RTF	MMS-1B-all	RTF 0.044 (weak accuracy — not production)

Top of leaderboard

Rank	Model	Robot	Acc%	WER	RTF	VRAM MB
1	Whisper-Large-v3-Turbo-CT2	89.62	68.40	0.316	0.046	2492
2	Whisper-Large-v3-CT2	87.92	72.92	0.271	0.133	4316
3	Whisper-Small-CT2	86.19	63.19	0.368	0.084	1340
4	Arabic-Whisper-Turbo-FT-CT2	83.38	60.76	0.392	0.046	1692
5	Arabic-Whisper-Large-v3-FT-CT2	79.11	63.19	0.368	0.135	2812
6	Voxtral-Mini-3B	60.19	71.53	0.285	0.281	11154

Takeaways

- Whisper CT2 (faster-whisper) models dominate the top of the robot leaderboard.
- Arabic fine-tunes were competitive on speed but did **not** beat base Whisper Large on accuracy in this bake-off.
- Voxtral-Mini is accurate but **~11 GB VRAM** — poor co-resident with LLM+TTS.
- Qwen ASR / QwenCleo underperformed on WER for this Egyptian clip — avoid for production.

3. LLM summary

4 models completed the prompt suite successfully (Nile-Chat-4B, Qwen3-4B-Instruct-2507, Qwen3-8B int4, ALLaM-7B). Automated scores cover **latency / throughput / VRAM**; Arabic answer quality was reviewed separately.

Recommended picks

Role	Model	Key metrics
Production default (speed UX)	Nile-Chat-4B	Robot 95.41 , TTFT 0.96 s , 11.56 tok/s , VRAM ~8.4 GB
Fastest TTFT	Qwen3-4B-Instruct-2507	TTFT 0.95 s , Quality ~4.41/5 , auto-pass ~80%
Quality alternative	Qwen3-8B (int4)	Quality ~4.65/5 , auto-pass ~85% , TTFT 1.94 s , VRAM ~6.9 GB

Role	Model	Key metrics
Avoid on ≤16 GB	ALLaM-7B	VRAM ~ 15.3 GB , slow TTFT 2.47 s

Leaderboard

Rank	Model	Robot	TTFT s	tok/s	VRAM MB	Quality /5	Auto-pass
1	Nile-Chat-4B	95.41	0.96	11.56	8394	~2.81	~20%
2	Qwen3-4B-Instruct-2507	91.61	0.95	10.48	8758	~4.41	~80%
3	Qwen3-8B (int4)	30.66	1.94	5.16	6900	~4.65	~85%
4	ALLaM-7B	4.15	2.47	5.62	15280	~4.25	~85%

Decision rule

- Prefer **Nile-Chat-4B** when turn-taking feel / low latency matters.
- Switch to **Qwen3-8B** (or Qwen3-4B) when answer correctness / TTS-fit matter more.
- Nile can be verbose — spot-check before freezing if TTS latency is tight.

4. TTS summary

4 models completed synthesis. Automated scores cover **speed / resources only** — Egyptian dialect and code-switch quality must be confirmed by **listening to WAVs**.

Recommended picks

Role	Model	Key metrics
Production default	VoiceTut-TTS	Robot 90.02 , RTF 0.157 , 82.7 chars/s , VRAM ~ 3.15 GB
Realtime alternative	SILMA-TTS	Robot 77.42 , RTF 0.238 , VRAM ~ 3.34 GB
Not realtime	Chatterbox / NAMAA	RTF > 1 — too slow for live robot

Leaderboard

Rank	Model	Robot	RTF	chars/s	VRAM MB	Realtime?
1	VoiceTut-TTS	90.02	0.157	82.7	3150	yes
2	SILMA-TTS	77.42	0.238	46.8	3340	yes
3	Chatterbox-Multilingual-V3	28.63	1.101	13.4	4790	no
4	NAMAA-Egyptian-TTS	0.61	1.232	15.1	6066	no

Listening checklist (manual)

Score each WAV 1–5 for: Egyptian naturalness, Arabic/English code-switch, numbers/dates, prosody/artifacts, overall robot-voice fit.

5. Combined families (stack budget)

Stack	ASR	LLM	TTS	Naive peak	Fits
Speed (default)	Turbo ~2.5 GB	Nile ~8.4 GB	VoiceTut ~3.2 GB	~14.0 GB	T4 tight / 24 GB OK
Quality	Large-v3 ~4.3 GB	Qwen3-8B ~6.9 GB	VoiceTut ~3.2 GB	~14.4 GB	T4 tight / 24 GB OK
Avoid	+ ALLaM ~15 GB	—	—	overflows mid GPUs	need ≥24–48 GB

Rule: ship on ≥24 GB even if T4 “fits,” so KV-cache, CUDA context, and queued turns do not OOM.

6. VPS / GPU hosting summary

Primary pick: RunPod Secure — RTX 4090 24 GB

Criterion	Why
Fit	24 GB hosts ASR+LLM+TTS with headroom
Interaction	SSH/root, Docker templates, REST API, serverless
Billing	Per-second; stop pod → stop compute
Reliability	Datacenter Secure tier (~99.9%+ class) vs Community marketplace

Practical rule: develop on **Community** (~\$0.34/hr) → go live on **Secure** (~\$0.69/hr).

Provider scorecard (July 2026 listings — re-check before buy)

Provider	RTX 4090-class \$/hr	Best use	Score /10
RunPod Secure	~0.69	Production default	9.0
RunPod Community	~0.34	Dev / staging	8.2
Vast.ai	~0.27–0.37	Cheap experiments	6.5
Hetzner	~0.33 (Ada 20 GB)	Always-on EU	7.8
DigitalOcean	RTX 4000 ~0.76	Simple DX	7.0
AWS / GCP / Azure	2–3× rates	Compliance only	4.5

Cost snapshots (1× RTX 4090)

Usage	RunPod Community	RunPod Secure	Vast.ai (~\$0.30)
3 h/day metered	~\$31	~\$62	~\$27

Usage	RunPod Community	RunPod Secure	Vast.ai (~\$0.30)
Business hours (10×22)	~\$75	~\$152	~\$66
Soft launch 24/7	~\$245	~\$497	~\$216
Growth (2 GPUs 24/7)	~\$490	~\$994	~\$432

Capacity (production speech lengths)

Assumptions: ~60 s user utterance, ~35 s robot reply, multi-turn sessions.

Stage	DAU	Turns/day	GPUs (24 GB) guidance
Pilot	100	~600	1
Soft launch	1,000	~8,000	1–2
Growth	5,000	~50,000	2–6
Scale	25,000	~300,000	8–12+ / serverless

One RTX 4090 $\approx$ **20–40** connected VAD sessions, **1–3** GPU-busy turns at once. Size from **peak concurrency**, not DAU alone.

Suggested architecture

Clients → Edge CPU (TLS/auth) → RunPod Secure RTX 4090
(Whisper Turbo + Nile/Qwen + VoiceTut)
→ Object storage (logs / analytics)

7. Final selection matrix

Decision	Choice
Default ASR	Whisper-Large-v3-Turbo-CT2
Accuracy ASR alt	Whisper-Large-v3-CT2
Default LLM	Nile-Chat-4B (speed UX)
Quality LLM alt	Qwen3-8B int4
Default TTS	VoiceTut-TTS (confirm by listening)
TTS alt	SILMA-TTS
GPU class	24 GB (RTX 4090 or better)
Provider	RunPod Secure Cloud
Dev / cost lane	RunPod Community or Vast.ai

8. Open sign-off items

1. Re-validate ASR on the real mic / noise / dialect mix (single-clip bake-off).
 2. Listen to VoiceTut (and SILMA) WAVs for Egyptian + code-switch quality.
 3. Spot-check Nile verbosity vs Qwen quality for the live UX priority.
 4. Re-check live GPU prices before committing budget.
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Summary of the Arabic Robot Research deliverables. Full detail remains in the individual PDF reports in this folder.

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